

“Get Real” Defining, Designing and Executing Virtual Production Curriculum at SCAD

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ABSTRACT

Higher educational programs in Savannah College of Art and Design’s (SCAD) School of Digital Media and Entertainment Arts have taken the leap to create a curriculum around virtual production including its adoption of the 40’x20’ LED volume set on SCAD’s new 10,000 square feet backlot for the Savannah Film Studios and aggressively including real-time training for its students. In this article, we summarize a new way to define, design and execute virtual production curriculum through lessons learned in its flagship production, “Get Real”.

KEYWORDS

virtual production, real-time, visual effects, LED volume, curriculum design, education

ACM Reference Format:

Dan Bartlett and Suanne Fu. 2022. “Get Real” Defining, Designing and Executing Virtual Production Curriculum at SCAD. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator’s Forum (SIGGRAPH ’22 Educator’s Forum)*, August 07–11, 2022, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532724.3538504>

1 INTRODUCTION

As December 2020’s pandemic dust settled onto higher education for remote teaching across the nation, the final month of the year offered a respite to look forward to new technologies and engagement for student learning. Reviewing the past year’s projects from SCAD’s School of Digital Media and SCADPro innovation studios, faculty from the programs of interactive design and game development, visual effects and motion media gathered to create a task force on putting forth a curriculum and proposals for the upcoming years for technology procurement. It was apparent that virtual production and LED volumes were a heavy interest to the future of the school. Initial proposals were simply to upgrade the current green screen rooms to a small LED wall and allow camera tracking, but SCAD had greater plans across the campus in School of Entertainment Arts to create a 10,000 square foot backlot. As 2021 opened, simple plans became bigger realities to catapult the programs of both schools into the future of virtual production (Figure 1).

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SIGGRAPH ’22 Educator’s Forum, August 07–11, 2022, Vancouver, BC, Canada

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ACM ISBN 978-1-4503-9366-9/22/08.

<https://doi.org/10.1145/3532724.3538504>



Figure 1: The school’s 50’x20’ LED volume (XR Stage) nestled in its new home amid SCAD’s new 10,000 square ft backlot of the SCAD Savannah Film Studios.

“GET REAL” – Scaffolding inaugural learning designs through a collaborative simulated production “Get Real”, the three-minute short film produced as the outcome of this collaboration between the Schools of Digital Media and Entertainment Arts, was in reality the second active production to utilize the physical and digital resources procured for SCAD’s virtual production facilities. Taking direct guidance from the methodologies used within the interactive design and game development program at SCAD, it made sense to take an iterative approach for the development of SCAD’s virtual production curriculum. The technical resources and shooting facilities (XR Stage) were launched in September 2021, SCAD’s Fall 2021 quarter. The Fall quarter attracted the initial cohort of students from the targeted majors, who were then formed into teams and set the task of producing a series of twenty-second audio-visual sequences that utilized the LED volume (XR Stage). Teams were given complete freedom to create whatever virtual environments best aligned to their project ideas. This explorative and arguably lax approach to the project parameters had a twofold advantage: first, it provided great opportunities to stress test a wider spectrum of individual components of the new technical resources; and, secondly, from a pedagogical perspective, allowed faculty to gauge the ease of which technical and creative skillsets from different majors were able to translate their knowledge and creative practice into a virtual production pipeline. From the schools of Entertainment Art and Digital Media, students from the following majors were brought together to participate in the first virtual production classes utilizing the new XR Stage at SCAD’s Savannah Film Studios (Figure 2).

This Fall 2021 “soft launch” project was successful in producing some high-quality visual artefacts and provided faculty with a clear picture of the technical limitations of the system, while also identifying the fundamental components that will be vital to the construction of a functional academic language and curriculum for



Figure 2: Live set for "Get Real" on the XR Stage.

virtual production. A key takeaway that would form the cornerstone for the "Get Real" project was that a combination of tighter creative scaffolding and looser technical parameters would produce the most inductive environment for exceptional project outcomes. Instructional instincts may naturally lean toward tighter technical parameters to control the results but in practice, our approach led to the student teams for "Get Real" the flexibility to explore more creatively, which eventually provided a complete, detailed screenplay, with senior academic administration acting as the client. The loosening of technical parameter was applied by firstly allowing the directors, cinematographers, and production design leads for each sequence a far greater wealth of access to the technical variables for on-set shooting (i.e., full lens kits for the Red Komodo camera kit and a full industry-standard lighting set-up; customizable to the needs of each unit director).

Despite the presence and active adoption of an institute-wide guide to pedagogical "best practice," individual departments have over time reduced the broader best practices down towards more concise, optimized pedagogical toolkits - specific to their respective disciplines. Immediately evident was the need to completely rethink the established communication strategies that were already embedded within each of the separate disciplines. Cinematographers and Game Artists do not have the same communication styles! But as an interesting parallel, Game Artists and on-set Production Designers DO share communication styles/preferences.

2 TEACHING PROJECT MANAGEMENT PRACTICES FOR VIRTUAL PRODUCTION

The necessary structure of the virtual production pipeline is a substantial shift from the production workflow that is effectively engrained within most film programs. This meant that our student producer (the amazing Sean Hussey) was the first person who needed to shift their working pattern, communication tools, and most importantly their preconceived assumptions. For our producers, their responsibility was to effectively manage and execute two parallel production workflows - virtual and physical. Each scene within the production had its own director, DP, and production designer. This structure was an active design decision by the faculty. Each production team was also working alongside a dedicated art and real-time effects team. Breaking the project down into smaller productions provided two immediate advantages. Firstly - smaller shooting sessions with multiple directors kept the project from feeling overwhelming for the student teams. This was the first active

production to use the LED volume at SCAD's new Savannah Film Studios, making the project very high-profile within the college. Secondly, not losing sight of the reality that these are student teams working within the confines of an academic ecosystem, smaller teams meant that should a single team's project stall or break down, the impact to the wider production could be minimized.

An immediately evident impact to this production structure was the additional responsibilities placed on the producers and creative leads. Four different production units for a short film is relatively uncommon within professional projects; four different production units for a film with a less than five-minute runtime is even rarer. "Get Real's" creative leads also faced the additional challenge of having to monitor and manage the asset pipeline and the materials produced by each unit's virtual art department. This was a significantly different workload and, in many cases, entirely new territories for the creative leads, who were majority film majors. The challenging workflow was overcome thanks to the skills brought forward by students majoring in interactive design and game development, and immersive reality. Both majors placed distinct emphasis on the creation and management of design assets as part of their project pipelines. Students from these majors, and their already comprehensive understanding of game engine art, design, and development workflows, became invaluable both in pre-production and on-set in facilitating the realisation of each unit directors' vision.

3 CONCLUSION

One year has passed since the official public announcement that Savannah College of Art and Design would be bringing Virtual Production facilities to the new Savannah Film Studios site. In September 2021 SCAD's Savannah Film Studios "Building 5" was opened; featuring a 40'x16'x'20 curved 1.5mm pixel-pitch LED volume, with a Disguise>Unreal Engine workflow. In the two academic quarters that followed SCAD faculty and students have moved at incredible speed and tenacity. Through small-scale iterative film projects, they have stress-tested these new technical workflows, understanding the production and communication methodologies that are pivotal to the success of any virtual production project. These projects provided a testbed to which we have begun to define and then construct the language and pedagogy of virtual production education. "Get Real" is the first tangible outcome of that journey, but still represents SCAD's earnest steps to establishing thought leadership within this next generation of digital arts education.

There are many takeaways from this project, but amongst the most prominent is the paramount importance of acknowledging and embracing the convergence of technical and creative skillsets. The technical skills, creative vision, and comprehensive understanding of iterative project management provided by immersive and game development majors were crucial. Equally valuable was the eye for visual aesthetic and narrative direction provided by film and visual effects majors. Moving forward to the next series of SCAD virtual production projects, including SCADPro sponsored projects and multiple senior films. It is evident that no student from any one single major can master virtual production alone; only through a convergence of skillsets and truly collaborative approach in pre-production and on-set will we continue to see successful outcomes like "Get Real."